

Incoterms 2020 & Global Trade Cost Engine

Comprehensive Project Specifications, Architecture, and Automated Deployment Documentation

- React & Vite SPA
- Tailwind CSS
- GitHub Actions CI/CD
- Production Ready

1. Executive Summary & Project Purpose

The **Incoterms 2020 & Global Trade Cost Engine** is an advanced, production-grade web application engineered to calculate comprehensive landed import/export costs, assess risk transitions under official International Chamber of Commerce (ICC) standards, and navigate global trade compliance. Built for integration into professional logistics portfolios, this tool provides real-time data calculations across multiple freight modalities.

2. Core Features & Architecture

Incoterms Master Matrix
Interactive reference breaking down all 11 standardized trade clauses, clarifying risk transfer points, cost allocations, and insurance duties.

Global Trade Cost Engine
Dynamic cost calculator accounting for FOB/CIF freight variables, multi-modal tariffs (Maritime, Air, Land), and currency conversions.

Compliance & Audits Hub
Structured compliance workflows, export document verification checklists, and audit trail modules to prevent regulatory delays.

Responsive React Architecture
High-performance Single Page Application (SPA) structured with modular components, local state management, and optimized asset bundles.

3. Technology Stack & Dependencies

Category	Technology / Library	Purpose
Core Framework	React (v18+) with Vite	High-speed UI rendering and lightning-fast module bundling.
Styling Engine	Tailwind CSS	Utility-first CSS framework for clean, modern responsive design.
Language	TypeScript / JavaScript	Type-safe robust application logic and data modeling.
Hosting & CI/CD	GitHub Pages & Actions	Automated build pipeline and free zero-cost public hosting.

4. Step-by-Step GitHub Actions Deployment Workflow

To establish continuous deployment from the repository to GitHub Pages without incurring costs, we configured an automated workflow using GitHub Actions. Below are the precise steps and configuration files utilized:

Step A: Configure Repository Settings

- Navigated to repository **Settings > Actions > General**.
- Updated **Workflow permissions** to *Read and write permissions* to allow the deployment script to commit built assets to the publishing branch.

Step B: Set Up Vite Base Path Configuration

Configured `vite.config.ts` to match the GitHub Pages subpath repository structure:

```
import { defineConfig } from 'vite' import react from '@vitejs/plugin-react' export default
defineConfig({ plugins: [react()], base: '/incoterms-trade-engine/', })
```

Step C: Automated CI/CD Workflow Script (`.github/workflows/deploy.yml`)

Implemented robust dependency resolution handling legacy peer dependencies (`--legacy-peer-deps`) to bypass strict npm conflict checks during build automation:

```
name: Deploy React App to GitHub Pages on: push: branches: [ main ] jobs: build-and-deploy:
runs-on: ubuntu-latest steps: - name: Checkout uses: actions/checkout@v4 - name: Set up Node.js
uses: actions/setup-node@v4 with: node-version: 20 - name: Install dependencies and build run: |
npm install --legacy-peer-deps npm run build - name: Deploy to GitHub Pages uses:
peaceiris/actions-gh-pages@v3 with: github_token: ${ secrets.GITHUB_TOKEN } publish_dir: ./dist
```

Production Status: Successfully compiled and live at <https://mohammed905-stack.github.io/incoterms-trade-engine/>